

Ruoshui (Walter) Li

Department of Mathematics and Statistics, Saint Louis University

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Academic portfolio: github.com/Walter-Li17/academic-projects

Education

Saint Louis University

Fall 2025 – Present

M.A. in Mathematics, expected Spring 2027

University of Connecticut

Sept. 2020 – Dec. 2024

B.A. in Mathematics

East China Normal University

Sept. 2020 – Dec. 2021

Study abroad program during the COVID-19 pandemic

Teaching Experience

Saint Louis University

Aug. 2026 – Present

Graduate Teaching Assistant; Pre-Calculus Instructor (MATH 1400, Fall 2026)

- Teach MATH 1400 Pre-Calculus, using worked examples, individual problem-solving, and small-group discussion.
- Design lecture notes, worksheets, quizzes, examinations, and solution keys; align weekly assessments with classroom practice.
- Use Canvas and MyMathLab to organize course materials and assignments; maintain a public teaching portfolio on GitHub.

Precalculus teaching and course-material preparation

Spring 2026

Prepared quizzes, review materials, examinations, and corresponding solutions.

Selected Academic Projects

Multivariable Taylor's Theorem

2026

Developed an exposition and presentation for MATH 5023 on directional derivatives and the multivariable Taylor theorem. Explained the proof by reduction to one variable and applications to error estimates and critical points.

Calculating Normal Probabilities without Tables

Fall 2025

Presented a graduate project for MATH 5021 at Saint Louis University on Bagby's 1995 approximation method, using double integrals and polar coordinates; prepared slides and a poster.

Research Interests

Artificial intelligence and machine learning, with a current focus on visual representations and image retrieval.

Technical Skills

Programming and ML: Python, NumPy, PyTorch, Hugging Face Transformers.

Methods: Pretrained vision-model inference, image feature extraction, retrieval evaluation, and numerical linear algebra (SVD and similarity measures).

Tools: LaTeX, GitHub, Canvas, and MyMathLab.